RM9: Approved separate-resource kernel and the strong-observation boundary

Method-first ramp-metering review

2 October 2026

Disposition. RM9-UG3.B is approved for analytic review. It supersedes only the original residual longitudinal-rate coupling. UG1.A priorities remain Default; other UG2 clauses and deferred X1–X3 remain. The approved amendment passes conditional conservation/storage checks and removes the universal lateral-rate debit. This report does not establish an exact equal-history/action pair, physical calibration or useful gradients. RM9-UG4 requests the first explicit speed/observation contract; no observer alternative below is adopted.

1 Approved update and conditional physical bounds

For two lanes per longitudinal cell, let $n_{i\ell}$ be vehicle-equivalent mass, L_i length, h timestep and $M_{i\ell}$ maximum mass. Write $c_{i\ell} = v_{i\ell}/L_i$, $\gamma_{i\ell} = w_{i\ell}/L_i$ and

$$D_{i\ell}(n) = \min(c_{i\ell}n, C_{i\ell}), \quad S_{i\ell}(n) = \min(C_{i\ell}, \gamma_{i\ell}(M_{i\ell} - n)). \tag{1}$$

With $hc_{i\ell} \le 1$, $h\gamma_{i\ell} \le 1$, both satisfy mass/storage step bounds. The adopted bounded propensity and actual transfer remain

$$A_{\ell j} = \mu_{\ell j} \max\left(0, \frac{\kappa_{\ell j}k_\ell - k_j}{\kappa_{\ell j}k_\ell + k_j}\right), \qquad \lambda_{\ell j} = \min(A_{\ell j}D_\ell(n_\ell), S_j(n_j)), \tag{2}$$

$$n_\ell^L = n_\ell + h(\lambda_{j\ell} - \lambda_{\ell j}), \qquad D_\ell^B = D_\ell(n_\ell^L), \quad S_\ell^B = S_\ell(n_\ell^L). \tag{3}$$

Here $k_\ell = n_\ell/L_i$, $\mu \in [0,1]$, $\kappa \in (0,1]$; zero denominator means both lanes empty and $A = 0$. At most one direction is positive. No original-budget-minus-lateral cap remains in the current law.

Ordinary forward flows use $q_{i\ell} = \min(D_{i\ell}^B, S_{i+1,\ell}^B)$. Where ramp and mainline share a forward receiver, retain the approved positive-priority, unused-supply allocator, with mainline share $a \in (0,1)$:

$$r = \min(d_r, \max((1-a)S^B, S^B - d_m)), \qquad m = \min(d_m, S^B - r). \tag{4}$$

It respects each offer and $m + r = \min(S^B, d_m + d_r)$. Meter permission limits upstream release; stored post-meter traffic can make downstream d_r different.

The lateral stage preserves lane storage and their total. If total accepted forward/ramp incoming rate $f_{i\ell} \le S_{i\ell}^B$ and outgoing $q_{i\ell} \le D_{i\ell}^B$, then

$$n_{i\ell}^+ = n_{i\ell}^L + h(f_{i\ell} - q_{i\ell}) \in [0, M_{i\ell}]. \tag{5}$$

Paired internal transfers cancel in the mass sum. Ramp storage, waiting and boundary transfers must obey corresponding balances; their exact geometry remains unspecified. This is a conditional kernel proof, not a completed network/ramp configuration or physical-fidelity validation.

2 The old debit is removed; useful control is still an open claim

On an ordinary interface with uncapped pre/post sending and ample downstream receiving, let lateral rate λ move from donor d to recipient t. The recomputed offers give

$$Q_B = c_d(n_d - h\lambda) + c_t(n_t + h\lambda) = Q_0 + h(c_t - c_d)\lambda, \quad Q_0 = c_d n_d + c_t n_t. \tag{6}$$

Thus B has no universal $-\lambda$ term. The recipient's sending can increase; its physical capacity does not automatically increase. Actual receiving or capacity limits can prevent any realized flow increase. At fixed cell length and state, this step-local rate correction tends to zero as $h \to 0$; the underlying continuous occupancy redistribution does not disappear. This is not a global convergence theorem or proof that maneuvers are costless.

For the earlier hand example $L = 1$, $h = 1/4$, $(c_d, c_t) = (1, 2)$, $(n_d, n_t) = (6, 2)$, $\mu = 1/2$, $\kappa = 1$, $\lambda = 3/2$:

$$(n_d^L, n_t^L) = (45/8, 19/8), \quad (q_d, q_t) = (45/8, 19/4), \quad Q_B = 83/8 = 10 + 3/8. \tag{7}$$

The original residual kernel gave $17/2$ instead. This compares two local operators on one input, not equal observable histories or a metering experiment.

Sensitivity path. On strict active branches away from ties and zero denominators, the lateral map, ordinary sending/receiving and priority allocator compose differentiably. For a full conserved state $x^{s+1} = F_B(x^s, \theta^s)$, its directional linearization is

$$z^{s+1} = F_{B,x} z^s + F_{B,\theta} \eta^s. \tag{8}$$

This is a conceptual/analytic path. If the meter is demand- or storage-limited, its local action derivative may vanish. A saturated, continuously busy shared receiver can redistribute stream service without changing total departures. No smoothing, numerical gradient check or independent usefulness is inferred.

Let P^s count *all* affected traffic still in the system, including external waiting, ramp storage and post-meter traffic. Under common arrivals A^s and final exits E^s,

$$P^{s+1} = P^s + h(A^s - E^s), \qquad J_h = h \sum_{s=0}^{T-1} P^s. \tag{9}$$

For a fixed common clearance horizon, fixed arrivals/initial mass and strict branches,

$$\partial_\theta J_h = -h^2 \sum_{s=0}^{T-2} (T - 1 - s) \partial_\theta E^s. \tag{10}$$

This accounting derivative is not a verified simulator gradient. If final departure histories agree, complete time agrees. A local merge-flow difference is not automatically a final-exit or useful-action difference.

3 What aggregate observations can recover on a free-flow branch

Distinguish the stored start-of-step state n^s, post-lateral state $n^{L,s}$, forward flows q^s, and end state n^{s+1}. Do not apply the old residual-kernel reconstruction to B: the observation law has changed. For an ordinary unimpeded interface under B,

$$q_\ell^s = c_\ell n_\ell^{L,s}. \tag{11}$$

Because lateral exchange conserves the cell total, $N^s = n_1^s + n_2^s = n_1^{L,s} + n_2^{L,s}$. With known distinct mobilities,

$$n_1^{L,s} = \frac{Q^s - c_2 N^s}{c_1 - c_2}, \qquad n_2^{L,s} = \frac{c_1 N^s - Q^s}{c_1 - c_2}. \tag{12}$$

An independent passage-speed observation remains admitted. Under the additional free-flow diagnostic that lane passage speeds equal distinct known v_ℓ,

$$V^s = \frac{v_1 q_1^s + v_2 q_2^s}{Q^s}, \qquad q_1^s = \frac{Q^s(V^s - v_2)}{v_1 - v_2}, \quad q_2^s = Q^s - q_1^s. \tag{13}$$

Equation (12) needs total and flow only on this subcase; it does *not* discard or redefine independently observed speed elsewhere. No equal-speed symmetry or noisy inversion is used.

A completed previous step can identify the current state. If the declared prior observations identify $n^{L,s}$ and all actual per-lane incoming/outgoing rates, then

$$n_\ell^{s+1} = n_\ell^{L,s} + h(f_\ell^s - q_\ell^s) \tag{14}$$

identifies the current state. Ramp entry must be observed or deduced from an audited ramp balance with common release/storage; unknown ramp mass cannot be silently ignored. These conditions can hold in an idealized slack/free-flow subcase. They do not cover a binding merge, missing stations or the entire congested domain. This is a conditional prior-history sufficiency result, not permission to observe a candidate action's future flow.

Where the proof stops. If $D(n^L) = C$ or downstream receiving/priority restricts flow, q/c need not equal n^L. Knowing total lane flow is then not enough to apply the free-flow inverse. Independent speed and full past history may still identify occupancy; their actual measurement functions and support are needed to decide. The first-order mass/flow kernel alone has not specified that detector-speed statistic. Snapshot non-uniqueness is not an equal-history/action certificate.

4 A stage ambiguity that is not useful hidden information

Even exact identification of n^L does not universally identify the preceding n under the accepted step bounds. For a source-limited lateral branch with $\kappa = \mu = 1$, fixed total N and donor lane 1 mass $x > N/2$,

$$\lambda(x) = c_1 x(2x - N)/N, \qquad \Phi(x) = x - h\lambda(x) = x(1 + b) - 2bx^2/N, \quad b = hc_1. \tag{15}$$

Its slope is $1 + b - 4bx/N$, which need not remain positive when $b > 1/3$. Mass/storage CFL bounds alone do not establish injectivity. This is not a violation of the bounds or automatic evidence of physical failure.

Exact hand counterexample. Take $N = 10$, $h = 1/2$, $(c_1, c_2) = (1, 3/2)$, $\kappa = \mu = 1$. Two strictly source-limited initial splits give

$$(n_1, n_2) = (6, 4), \qquad \lambda = 6/5, \tag{16}$$

$$(n_1', n_2') = (9, 1), \qquad \lambda' = 36/5. \tag{17}$$

Both become

$$(n_1^L, n_2^L) = (27/5, 23/5). \tag{18}$$

Finite compatible triangular-diagram parameters can be $(M_1, M_2) = (100, 100)$, $(\gamma_1, \gamma_2) = (1, 3/2)$ and $(C_1, C_2) = (50, 75)$, with ample downstream receiving. All step bounds hold, both states are strictly free-flowing and recipient supplies exceed transfer rates. No symmetry of lane mobilities is imposed.

If all remaining states/forcing agree and the candidate control acts only in the forward/ramp stage after this control-independent lateral map, these two pre-states produce identical responses to *every* such action. Their post-transfer states are the same; the subsequent deterministic composition is therefore the same. No useful action distinction follows from this lost pre-stage coordinate. Earlier observable history may also distinguish them; no such history match has been established.

This prevents two mistaken conclusions: that a post-transfer observer theorem automatically reconstructs the pre-transfer state, or that failure to reconstruct that pre-state automatically proves useful hidden information. Do not restrict the approved propensity or timestep simply to make an inverse proof succeed. A monotonicity/refinement requirement, if later needed for physical fidelity, is a separate declared amendment rather than a hidden fix.

5 RM9-UG4: define speed, support and measurement timing

Problem. Which strong observation map is used to decide whether lane information is hidden? The approved dynamics specify mass and accepted flow, but not an independent detector passage-speed statistic for congested or receiving-limited traffic. Density, flow and speed can refer to different locations, averaging periods and update stages.

Model-chain function and core gap. Completed observations, common forcing and past rates inform the decision; the common rate perturbation then propagates through the approved dynamics to all-stream departures/time. Choosing a speed map or moving a measurement across the lateral stage can change identifiability. It cannot be inferred as clerical transcription. No future candidate-action flow may enter its initial observation history.

A: explicit idealized first-order observation benchmark (recommended). Keep true road-total mass per declared cell, ramp history, counts, all past rates and known forcing from the common initial epoch. Report flow and speed for completed steps at explicitly declared faces, preserving their stage indices. Propose lane-local equilibrium FD speed $U_\ell(k)$ and a separate passage-weighted road statistic:

$$U_\ell(k) = \frac{\min(v_\ell k, C_\ell, w_\ell(k_\ell^{\mathrm{jam}} - k))}{k}, \quad k > 0, \qquad V = \frac{\sum_\ell q_\ell U_\ell(k_\ell^L)}{\sum_\ell q_\ell}. \tag{19}$$

This assumes a single representative equilibrium speed within each lane and an explicit upstream-cell-to-face passage approximation. It is an observation hypothesis, not a detector theorem. At zero observed flow, passage speed is unavailable; do not fabricate an informative value. On free flow this reduces to the diagnostic already proved. On congested/transient faces it requires explicit source/model justification and may fail to represent actual passage speeds. No aggregate Q/K replacement or speed-state extension is adopted.

Pros: concrete, compact, retains the independent speed moment and the strong total-state/history benchmark; supports bounded exact information analysis without new traffic dynamics. *Cons:* idealized cell/station and within-lane speed assumptions; a result is conditional on this observer, not field validity. *Dependencies:* approve this hypothesis before exact history/action construction; specify complete temporal support, ramp balances and common geometry consistently. If the passage approximation fails, report/reopen it.

B: source/measurement-first speed-contract review. Keep the accepted strong information requirements, but perform a targeted static review of station/window passage speeds and their relation to independent road-total density before choosing a map. Do not substitute detector-derived density for true cell storage or adopt extra speed dynamics automatically.

Pros: grounds the measured statistic and transient/congested support before exact information claims; can identify a required richer speed model. *Cons:* delays the compact analytic test; may require a separately approved measurement/state amendment or leave the mapping unresolved. *Dependencies:* bound the source review and stop before adding state, weakening information or executing a simulator. Existing first-order results remain conditional diagnostics.

Provisional recommendation: A for the first idealized methodology benchmark, with its observer hypothesis explicit and physical transfer claims deferred. Choose B if physical detector interpretation must be secured first. This is a G1 contract choice, not a numerical-cost or parameter-value gate.

Source boundary. FB0030's primary PDF, printed pp. 1768–1769, distinguishes time/space means and uses detector flow/speed to estimate density; this does not provide independent true cell mass and our exact transient passage map. RM0018's primary PDF supports the retained node allocator, not this speed map. These limitations were rechecked in primary PDFs. All new results here are reviewer derivations, not source theorems.